a) $Y[n] = \cos(X[n]) + n$

Linearity 1)	X
Time-Invarience 2)	X
Causality 3)	✓
Memory-less 4)	✓

1) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = \cos(x'[n]) + n = \cos(\alpha x[n]) + n \neq \alpha y[n]$$

2) $x'[n] = x[n+d]$

$$y'[n] = \cos(x'[n]) + n = \cos(x[n+d]) + n \neq y[n+d] = \cos(x[n+d]) + n + d$$

b) $Y[n] = 2^{-n} \cdot x[n]$

1) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = 2^{-n} \cdot x'[n] = 2^{-n} \cdot \alpha x[n] = \alpha y[n]$$

2) additivity: $x'[n] = x_1[n] + x_2[n]$

$$y'[n] = 2^{-n} \cdot x'[n] = 2^{-n} \cdot (x_1[n] + x_2[n]) = y_1[n] + y_2[n]$$

1)	✓
2)	X
3)	✓
4)	✓

2) $x'[n] = x[n+d]$

$$y'[n] = 2^{-n} \cdot x'[n] = 2^{-n} \cdot x[n+d] \neq 2^{-n+d} \cdot x[n+d] = y[n+d]$$

c) $Y[n] = \sum_{k=0}^{n} 0.9^{k} \cdot x[n-k]$

1)	✓
2)	X
3)	✓
4)	X

1) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = \sum_{k=0}^{n} 0.9^{k} x'[n] = \sum_{k=0}^{n} 0.9^{k} \cdot \alpha \cdot x[n-k] = \alpha \cdot \sum_{k=0}^{n} 0.9^{k} \cdot x[n-k] = \alpha Y[n]$$

2) additivity: $x'[n] = x_1[n] + x_2[n]$

$$y'[n] = \sum_{k=0}^{n} 0.9^{k} x'[n-k] = \sum_{k=0}^{n} 0.9^{k} \cdot (x_1[n-k] + x_2[n-k]) = Y_1[n] + Y_2[n]$$

2) $x'[n] = x[n+d]$

$$y'[n] = \sum_{k=0}^{n} 0.9^{k} x'[n-k] = \sum_{k=0}^{n} 0.9^{k} x[n-k+d] \neq$$

$$\neq \sum_{k=0}^{n+d} 0.9^{k} x[n-k+d] = Y[n+d]$$

d) $Y[n] = \dfrac{x[n]}{1 + x[n]}$

1)	X
2)	✓
3)	✓
4)	✓

1) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = \frac{x'[n]}{1 + x'[n]} = \frac{\alpha x[n]}{1 + \alpha x[n]} \neq \alpha Y[n]$$

2) $x'[n] = x[n+d]$

$$y'[n] = \frac{x'[n]}{1 + x'[n]} = \frac{x[n+d]}{1 + x[n+d]} = Y[n+d]$$

e) $Y[n] = (x[n])^{3}$

1	X
2	✓
3	✓
4	✓

1) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = (x'[n])^{3} = \alpha^{3} x[n]^{3} \neq \alpha Y[n]$$

2) $x'[n] = x[n+d]$

$$y'[n] = (x'[n])^{3} = x^{3}[n+d] = Y[n+d]$$

f) $Y[n] = X[n] - X[n-1] + 0.5\,Y[n-1]$

$$Y[n] = X[n] - X[n-1] + 0.5\left(X[n-1] - X[n-2] + 0.5\,Y[n-2]\right)$$

$$Y[n] = X[n] - X[n-1] + 0.5 \cdot X[n-1] - 0.5 \cdot X[n-2] + 0.5\,Y[n-2]$$

$$Y[n] = \sum_{k=-\infty}^{0} 0.5^{-k} \cdot \left(X[n+k] - X[n+k-1]\right)$$

1	✓
2	✓
3	✓
4	X

1) homogeneity:

$$X'[n] = \alpha X[n]$$

$$Y'[n] = \sum_{k=-\infty}^{0} 0.5^{-k} \left(X'[n+k] - X'[n+k-1]\right) =$$

$$= \sum_{k=-\infty}^{0} 0.5^{-k} \left(\alpha X'[n+k] - \alpha X'[n+k-1]\right) =$$

$$= \alpha \sum_{k=-\infty}^{0} 0.5^{-k} \left(X[n+k] - X[n+k-1]\right) = \alpha Y[n]$$

additivity: $X'[n] = X_1[n] + X_2[n]$

$$Y'[n] = \dots = Y_1[n] + Y_2[n]$$

2) $X'[n] = X[n+d]$

$$Y'[n] = \sum_{k=-\infty}^{0} 0.5^{-k} \cdot \left(X[n-k+d] - X[n-k+d-1]\right) = Y[n+d]$$

g) $Y[n] = \begin{cases} X[n], & |X[n]| \leq 1 \\ 0, & \text{otherwise} \end{cases}$

1	X
2	X
3	✓
4	✓

1) $X'[n] = \alpha X[n]$

$\exists\, \alpha$ s.t. $\alpha \cdot X[n] > 1 \quad \forall X[n] \neq 0 \Rightarrow$

$\Rightarrow X'[n] = 0 \neq \alpha Y[n]$

2) $x'[n] = x[n+d] \Rightarrow y'[n] = \begin{cases} x'[n], & |x'[n]| \leq 1 \\ 0, & \text{otherwise} \end{cases} \Rightarrow$

$\Rightarrow y'[n] = \begin{cases} x[n+d], & x[n+d]| \leq 1 \\ 0, & \text{otherwise} \end{cases} = y[n+d]$

h) $y[n] = e^{x[n]} - 1$

1	x
2	✓
3	✓
4	✓

1) homogeneity: $x'[n] = \alpha x[n]$

$y'[n] = e^{x'[n]} - 1 = e^{\alpha x[n]} - 1 \neq \alpha \cdot y[n]$

2) $x'[n] = x[n+d]$

$y'[n] = e^{x'[n]} - 1 = e^{x[n+d]} - 1 = y[n+d]$